Applications: Growth and Decay

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Growth models

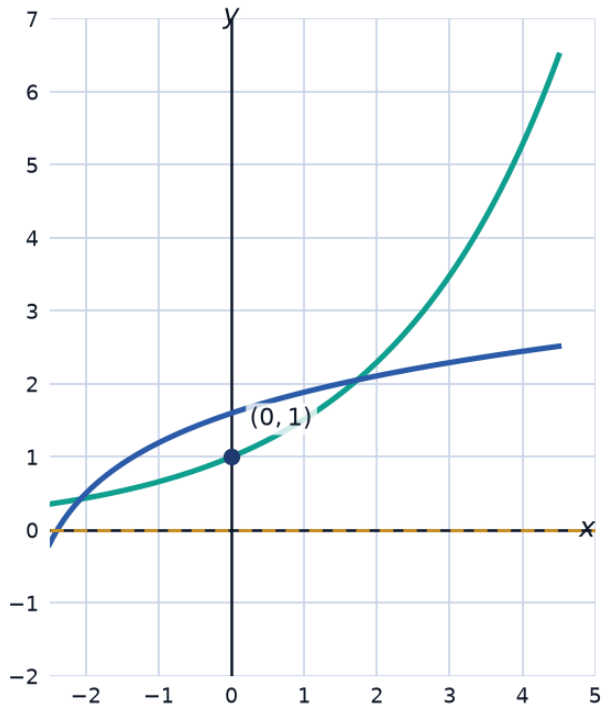
A city's population is modeled by $P(t) = 12000(1.035)^t$, where t is in years.

a State the initial population and the annual growth rate.

b Find the population after 10 years, to the nearest person.

2 [Foundational] • Target: Growth models

An account earns 4% annual interest compounded continuously. How long does it take an investment to double? Give an exact expression and a decimal to two places.



3 [Foundational] • Target: Decay models

Carbon-14 has a half-life of 5730 years: $A(t) = A_0\left(\frac{1}{2}\right)^{t/5730}$.

a What fraction of the original carbon-14 remains after 2000 years? Answer to three decimal places.

b A sample retains 30% of its carbon-14. Estimate its age to the nearest hundred years.

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4 [Foundational] • Target: Decay models

A 200 g sample of a radioactive substance decays to 50 g in 12 hours, following $A(t) = 200e^{kt}$.

a Find the decay constant k (4 d.p.).

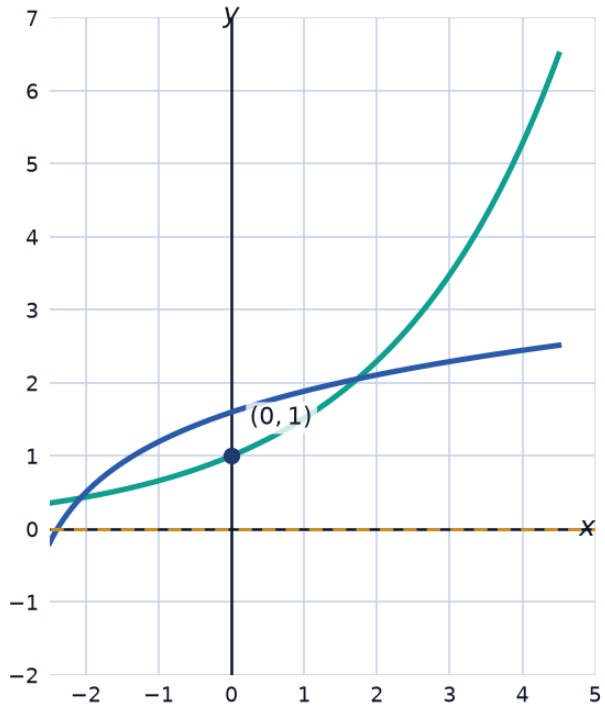
b Find the half-life of the substance.

5 [Foundational] • Target: Decay models

Newton's law of cooling: coffee at $90°$C cools in a $20°$C room following $T(t) = 20 + 70e^{-0.1t}$ (t in minutes). Find the temperature after 10 minutes, to one decimal place.

6 [Foundational] • Target: Logistic growth (an introduction)

A lake is stocked with fish, and the population is modeled by $P(t) = \frac{5000}{1 + 9e^{-0.4t}}$, where t is in years.



a Find the initial population $P(0)$.

b Find the carrying capacity (the value P approaches as $t \to \infty$).

c Find $P(10)$, to the nearest fish.

7 [Foundational] • Target: Comparing models

Two towns start with 10,000 people. Town A grows at a constant rate of 300 people/year (linear); Town B grows at 3% per year (exponential). Find each town's population after 15 years, to the nearest person, and state which is larger.

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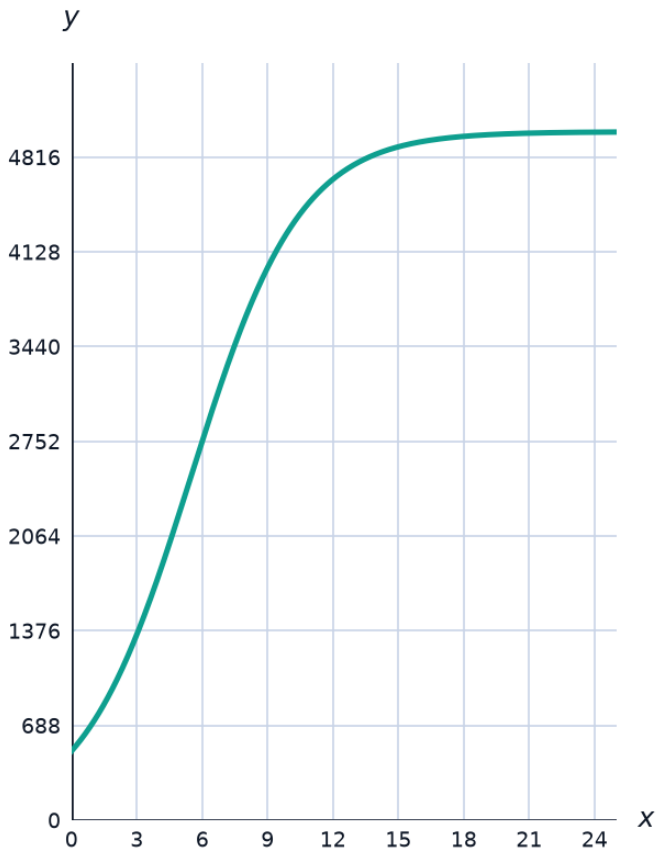
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Comparing models**

A radioactive isotope decays from 80 g to 10 g over 3 half-lives. Find the half-life in terms of the decay time, and write the exponential decay model $A(t) = 80e^{kt}$, finding k if the total elapsed time is 15 years (4 d.p.).

9 **[Intermediate]** • **Target: Visualizing the logistic curve**

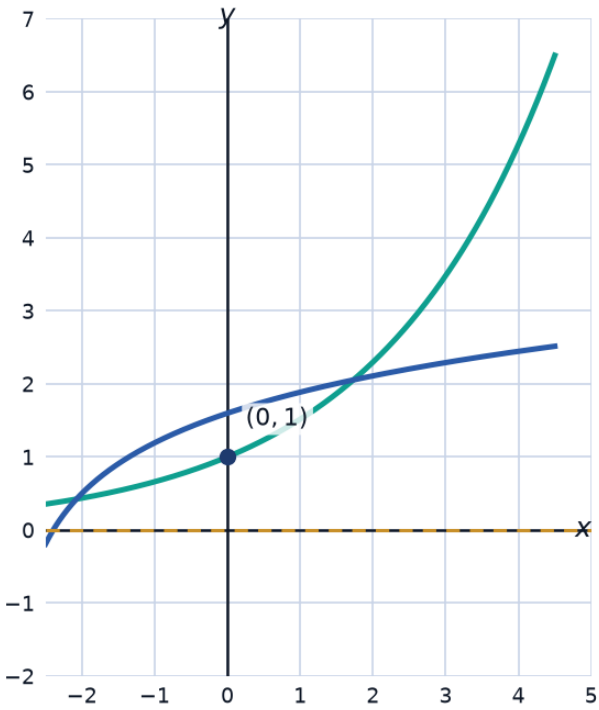
The fish population model $P(t) = \frac{5000}{1 + 9e^{-0.4t}}$ (from earlier) is graphed below. Use the graph to estimate where the population is growing fastest, then explain why this happens near the inflection point.



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10 [Intermediate] • Target: Doubling time and half-life formulas

Derive a general formula for the doubling time T of a quantity growing as $A(t) = A_0 e^{kt}$, in terms of k.



11 [Intermediate] • Target: Doubling time and half-life formulas

Using the doubling-time formula derived above, find the doubling time for a population growing at a continuous rate of $k = 0.025$ per year, to one decimal place.

12 [Intermediate] • Target: Half-life formula derivation

Derive a general formula for the half-life $T_{1/2}$ of a quantity decaying as $A(t) = A_0 e^{kt}$ (with $k < 0$), in terms of k.

13 [Intermediate] • Target: Half-life formula derivation

A radioactive isotope has decay constant $k = -0.023$ per year. Use the formula above to find its half-life, to one decimal place.

14 [Intermediate] • Target: Combining growth models with rates

A savings account earns continuous interest, growing from 5000 to 5512.50 riyals in 2 years. Find the annual interest rate r using $A = Pe^{rt}$, to two decimal places (as a percentage).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

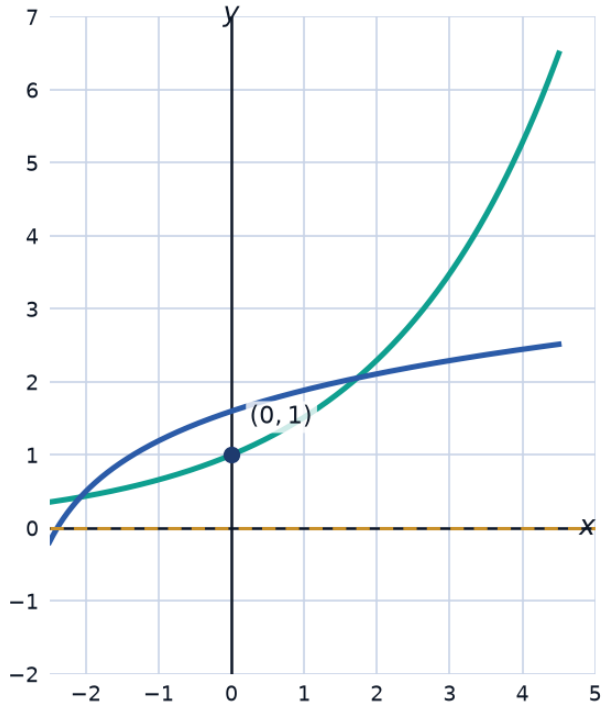
15 [Challenging] • Target: Logistic growth (an introduction) • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Solve $\cos(2x) + \sin x = 0$ on $[0, 2\pi)$. (attempt 10)

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16 [Challenging] • Target: Comparing models • Cross-Lesson: 3.03 Sum and Difference Formulas

Consider the exponential growth model $P(t) = P_0 e^{kt}$, where $P(t)$ is the population at time t and P_0 is the initial population. If the initial population is $P_0 = 500$ and after 10 years the population is $P(10) = 1200$, find the annual growth rate k. Round your answer to four decimal places.



17 [Challenging] • Target: Visualizing the logistic curve • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Consider the exponential growth model described by the function $P(t) = P_0 e^{kt}$, where $P(t)$ is the population at time t, P_0 is the initial population, and k is the growth constant. If the initial population P_0 is 500 and the population after 4 years is 2000, determine the value of the growth constant k to three decimal places.

18 [Challenging] • Target: Doubling time and half-life formulas • Cross-Lesson: 4.01 Circles and Parabolas

A population of bacteria, initially P_0 cells, grows according to the logistic model $P(t) = \frac{L}{1 + Ae^{-kt}}$, where t is in hours, L is the carrying capacity, and A and k are positive constants. At $t = 0$, the population is $P(0) = \frac{L}{3}$. After 2 hours, the population has doubled to $P(2) = \frac{2L}{3}$. Find the exact value of e^{2k} in terms of L and P_0.

19 [Challenging] • Target: Half-life formula derivation • Cross-Lesson: 4.02 Ellipses

The graph of the function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of the parameter a.

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20 [Challenging] • Target: Combining growth models with rates • Cross-Lesson: 4.03 Hyperbolas

Consider the exponential function $f(x) = 2^x$. If $f(a) = 8$, what is the value of a? Show your complete algebraic work.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Population Growth Models

A bacterial culture doubles every 3 hours. If the initial count is 500, what is the population after 9 hours?

a. 4000

b. 1500

c. 2000

d. 8000

22 [Foundational] • Target: Population Growth Models

A city has population $P(t) = 100000e^{0.02t}$, where t is in years. Find the continuous annual growth rate and doubling time.

a. Rate $= 2\%$, Doubling time ≈ 34.7 years

b. Rate $= 20\%$, Doubling time ≈ 3.5 years

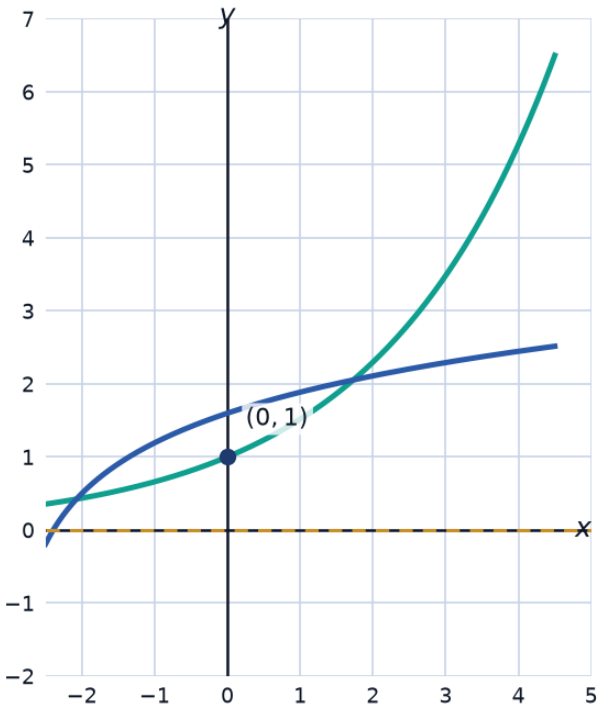
c. Rate $= 0.02\%$, Doubling time ≈ 50 years

d. Rate $= 2\%$, Doubling time ≈ 50 years

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23 [Foundational] • Target: Population Growth Models

If a culture grows from 200 to 600 in 4 hours according to $P(t) = P_0e^{kt}$, find k.



a. $k = \frac{\ln 3}{4} \approx 0.275$

b. $k = \frac{\ln 4}{3}$

c. $k = 3/4$

d. $k = \ln(3/4)$

24 [Foundational] • Target: Radioactive Decay and Half-Life

Carbon-14 has a half-life of 5730 years. What fraction of the original C-14 remains in a fossil after 17190 years?

a. 1/8

b. 1/4

c. 1/16

d. 1/3

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25 [Foundational] • Target: Radioactive Decay and Half-Life

State the formula relating decay constant $k > 0$ and half-life $t_{1/2}$ in $N(t) = N_0 e^{-kt}$.

a. $k = \frac{\ln 2}{t_{1/2}}$

b. $k = \frac{t_{1/2}}{\ln 2}$

c. $k = \frac{1}{2t_{1/2}}$

d. $k = \ln(2t_{1/2})$

26 [Foundational] • Target: Radioactive Decay and Half-Life

A sample contains 80 mg of an isotope with a half-life of 10 days. How much remains after 25 days?

a. $80(0.5)^{2.5} \approx 14.14$ mg

b. 20 mg

c. 10 mg

d. 15 mg

27 [Foundational] • Target: Newton's Law of Cooling

State Newton's Law of Cooling formula for temperature $T(t)$ with ambient medium T_m.

a. $T(t) = T_m + (T_0 - T_m)e^{-kt}$

b. $T(t) = T_0 e^{-kt} + T_m$

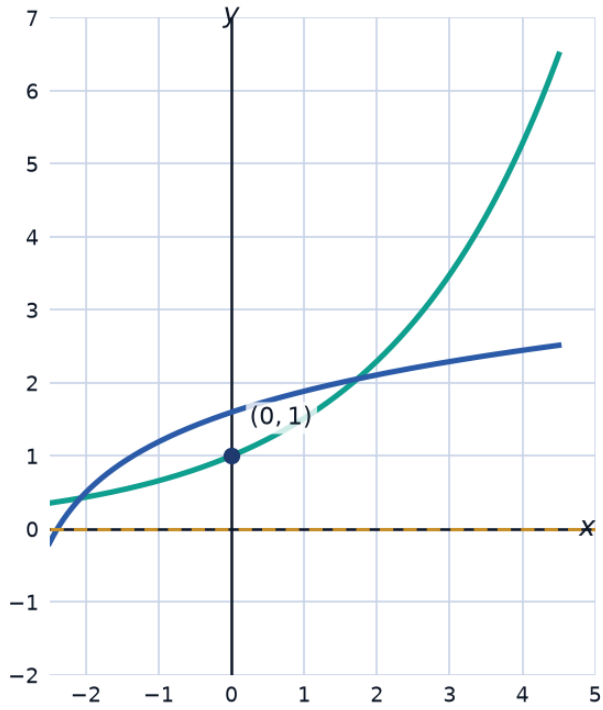
c. $T(t) = T_m(1 - e^{-kt})$

d. $T(t) = (T_0 - T_m)e^{kt}$

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28 [Foundational] • Target: Newton's Law of Cooling

A hot cup of coffee at 190°F is placed in a 70°F room. What is the initial temperature difference $T_0 - T_m$?



a. 120°F

b. 190°F

c. 70°F

d. 260°F

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Newton's Law of Cooling

If the coffee cools from 190°F to 130°F in 10 minutes in the 70°F room, find k.

a. $k = \frac{\ln 2}{10} \approx 0.0693 \text{ min}^{-1}$

b. $k = 0.1$

c. $k = \frac{\ln 3}{10}$

d. $k = 0.05$

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30 **[Intermediate]** • **Target: Logarithmic Scales (pH, Richter, Decibels)**

The pH of a solution is defined as $\text{pH} = -\log_{10}[\text{H}^+]$. What is the pH if $[\text{H}^+] = 10^{-4}$ M?

a. 4

b. -4

c. 10

d. 14

31 **[Intermediate]** • **Target: Logarithmic Scales (pH, Richter, Decibels)**

The Richter magnitude scale is $M = \log_{10}\left(\frac{I}{S}\right)$. How many times more intense is an earthquake of magnitude 7 than one of magnitude 4?

a. 1000 times (10^3)

b. 3 times

c. 30 times

d. 10000 times

32 **[Intermediate]** • **Target: Logarithmic Scales (pH, Richter, Decibels)**

Sound intensity level in decibels is $\beta = 10\log_{10}\left(\frac{I}{I_0}\right)$. If the intensity I is multiplied by 100, how many decibels are added?

a. +20 dB

b. +100 dB

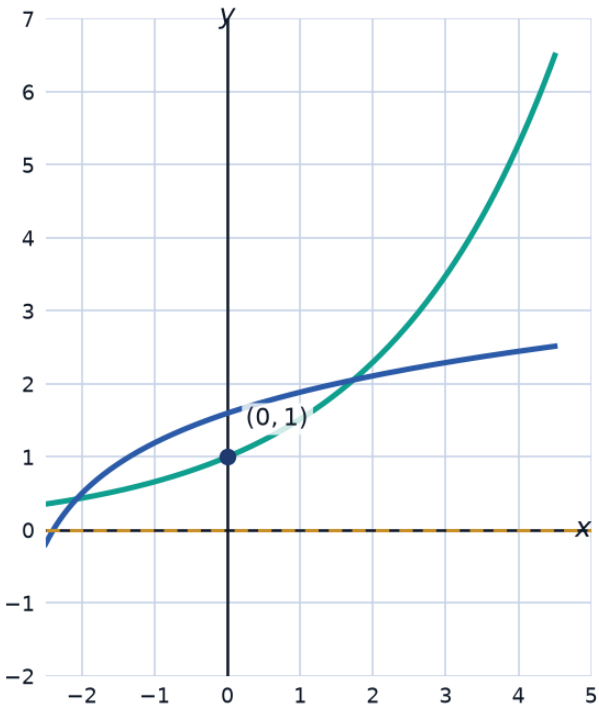
c. +10 dB

d. +2 dB

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33 [Intermediate] • Target: Growth models

Consider the exponential function $f(x) = 2^x$. What is the value of $f(3)$?



a. 6

b. 8

c. 9

d. 5

34 [Intermediate] • Target: Decay models

Consider the exponential function $f(x) = 3^x$. What is the value of $f(2)$?

a. 6

b. 9

c. 8

d. 5

35 [Intermediate] • Target: Logistic growth (an introduction)

A population of 100 grows at 2% per year, modeled by $P(t) = 100(1 + 0.02)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 100$)?

a. 22.5 years

b. 20.5 years

c. 26.7 years

d. 18.5 years

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36 [Intermediate] • Target: Comparing models

A population of 500 grows at 3% per year, modeled by $P(t) = 500(1 + 0.03)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 500$)?

a. 15.7 years

b. 17.8 years

c. 11.7 years

d. 13.7 years

37 [Intermediate] • Target: Visualizing the logistic curve

A population of 500 grows at 2% per year, modeled by $P(t) = 500(1 + 0.02)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 500$)?

a. 22.5 years

b. 20.5 years

c. 26.7 years

d. 18.5 years

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Doubling time and half-life formulas • Cross-Lesson: 4.01 Circles and Parabolas

A population of 1000 grows at 2% per year, modeled by $P(t) = 1000(1 + 0.02)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 1000$)?

a. 26.7 years

b. 18.5 years

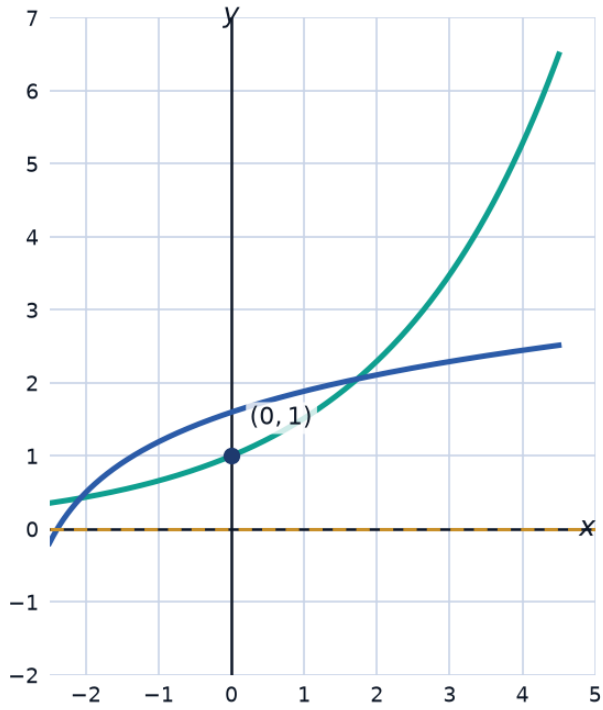
c. 22.5 years

d. 20.5 years

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39 [Challenging] • Target: Half-life formula derivation • Cross-Lesson: 4.02 Ellipses

A population of 200 grows at 2% per year, modeled by $P(t) = 200(1 + 0.02)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 200$)?



a. 22.5 years

b. 18.5 years

c. 26.7 years

d. 20.5 years

40 [Challenging] • Target: Combining growth models with rates • Cross-Lesson: 4.03 Hyperbolas

A population of 1000 grows at 3% per year, modeled by $P(t) = 1000(1 + 0.03)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 1000$)?

a. 17.8 years

b. 15.7 years

c. 13.7 years

d. 11.7 years

41 [Challenging] • Target: Population Growth Models • Cross-Lesson: 5.01 Vectors: Components and Operations

A population of 500 grows at 5% per year, modeled by $P(t) = 500(1 + 0.05)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 500$)?

a. 8.3 years

b. 10.8 years

c. 6.3 years

d. 10.3 years

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42 [Challenging] • Target: Radioactive Decay and Half-Life • Cross-Lesson: 5.02 The Dot Product

A population of 500 grows at 8% per year, modeled by $P(t) = 500(1 + 0.08)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 500$)?

a. 5.3 years

b. 7.3 years

c. 3.3 years

d. 6.9 years

43 [Challenging] • Target: Newton's Law of Cooling • Cross-Lesson: 5.03 Parametric Equations

A population of 200 grows at 8% per year, modeled by $P(t) = 200(1 + 0.08)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 200$)?

a. 5.3 years

b. 6.9 years

c. 7.3 years

d. 3.3 years

44 [Challenging] • Target: Logarithmic Scales (pH, Richter, Decibels) • Cross-Lesson: 6.01 Polar Coordinates and Conversions

A population of 200 grows at 5% per year, modeled by $P(t) = 200(1 + 0.05)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 200$)?

a. 10.3 years

b. 6.3 years

c. 10.8 years

d. 8.3 years

45 [Challenging] • Target: Growth models • Cross-Lesson: 6.02 Polar Equations and Graphs

A population of 100 grows at 5% per year, modeled by $P(t) = 100(1 + 0.05)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 100$)?

a. 6.3 years

b. 10.3 years

c. 8.3 years

d. 10.8 years

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